Estimating country literacy rates using a regression line based on country data

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1 Introduction

The pace of economic development of countries is based on people and their livelihoods. The choices that each individual makes in the allocation of his/her resources, activities at work, personal life, together make up contribution to the economy of the country. Among these factors, a separate place is occupied by the level of education. The extent to which the country's population is literate affects the pace of economic growth in the truest sense. Increased labor productivity, resulting from increased tax deduction, lower poverty and lower inequality are among the consequences of higher education, among many others. The study of the relationship between the level of literacy and economic factors is incredibly important for building the policy of the country's leadership, labor market research, and the fight against poverty and inequality. The Organization for Economic Cooperation Development recently released an article according to which, for the first time since the start of the PISA research, a significant decrease in literacy rates among 15-year-olds around the world has been recorded. The report highlights that the COVID-19 pandemic has significantly affected the quality of education, especially in low- and middle-income countries. In some European countries, such as Germany and France, there has also been a marked decline in results, which is causing concern among experts (OESD, 2023).

The purpose of the paper is to analyze what factors influence the literacy rate of the population of countries. The research aims to identify which factors affect it the least, thus, have little or no difference on the ability of the population to write or read (Kann and Taylor, 1988).

The modern period is characterized by a historically high literacy rate, which is about 86% among adults and 92% among young people (UNESCO, 2017). Nevertheless, Illiteracy remains a significant challenge globally, particularly in developing countries, where over 750 million adults lack basic literacy skills (Coulombe et al., 2004). Researching the relationship of economic factors and literacy rate can help explain the reasoning behind countries with low literacy rates facing slower GDP growth, higher poverty levels, and weaker competitiveness in the global market.

While studies in the past focused on GDP itself affecting the literacy rate of countries, this study aims to improve this notion by looking into the effect of GDP

per capita. As a result of the regression model, described in the paper, it was found that the GDP per capita has initially a negative effect up to a certain turnover point and a positive impact after that. Moreover, new variables such as population density were added and analyzed which has not been looked at before. The variable enrollment which was vague due to the fact that included all types of education was split into 2 in this research: tertiary (university) and secondary (high school) education to study their effect independently and more precisely.

2 Literature Review

The factors affecting the level of literacy of the country's population require careful analysis. Often, many of these factors are closely interrelated and influence each other, which is a separate problem in studying this issue. At the same time, it is difficult to represent individual factors in a numerical format for proper use in a regression model. Historical and geographical factors in countries also affect the determination of literacy levels. At the same time, there is a high probability of missing key indicators that have something to do with the model, which will undoubtedly affect the veracity of the results.

The multiplicity of factors and their interrelationship require careful analysis in order to take into account the context of each country or region and develop comprehensive strategies to improve the level of education.

The variables that we decided to use in building the regression model were chosen as a result of studying past studies. As a result, the following variables were selected: Our World in Data To determine the factors that influence literacy rate across countries, datasets from Our World in Data (OWID) were taken. OWID is a respected platform that provides global statistics, making it a valuable source for this project. It is managed by the Global Change Data Lab and supported by the University of Oxford, which highlights its credibility. The platform is widely used by academics, policymakers, and media outlets highlighting its reliability. Another reason for selecting OWID as a datasets' source is that during the econometrics course, this website was listed as one of the recommended sources for students' projects. This factor demonstrates the academic trust placed in OWID's data quality. Its datasets are sourced from reputable organizations such as UNESCO, the World Bank, and UN agencies. Most of the justification for the choice of the variables in our models that follows in the Methodology section of the paper was partly taken from Verner & Verner, 2005. This book discusses what influences the literacy rate of several countries and provides numerous factors. GDP per capita was also chosen as a major variable that directly affects the level of education and is based on the information taken from Rahman, 2011 .

3 Methodology

The objective of this study is to analyze the factors that influence the literacy rate of the population in different countries. The study is based on a cross-sectional data sample, including information on key factors from 110 countries. This section

describes in detail the process of data selection, the choice of variables for analysis, as well as the stages of data preparation and statistical analysis. As mentioned earlier, a significant part of the data was obtained from the Our World in Data web platform. The data from 2022 is used. The data set initially included 200 countries, but due to the lack of information for a number of countries, the sample size was reduced to 110 countries for which complete information was available for the analysis. This decision was made to maintain accuracy of the analysis, as incomplete data could have skewed the results.

Following variables in a regression model were selected from OWID:

Dependent Variable:

- **Cross-country literacy rates (LitRate):** This dataset serves as the dependent variable, representing each country's percentage of literate people.

Independent Variables:

- **Gross Enrollment Ratios in Secondary (high school) and Tertiary (university) Education (GERsec, GERtert):** The percentage of the eligible official school and university age population corresponding to the same level of education in a given year. The variable was chosen since higher level of enrollment in education is associated with higher levels of literacy.

- **Population Density (PopDens):** Demographic factors like population density can affect school access, especially in rural areas. The variable was chosen since higher population density is associated with a fewer opportunities to access education. To illustrate this, an example of countries with a large population that needs to be educated can be taken.

- **Share of individuals using the Internet (Internet):** Internet access is crucial for modern education, enabling access to information and online learning platforms. Higher access to the Internet is associated with a higher level of literacy of the population.

- **GDP Per Capita (GDPC) and GDP Per Capita Squared (GDPCsq):** GDP of a country that measures its economic well-being directly affects its literacy rate for the reason that it determines the amount of funds that go into the education sector of a country. To illustrate, after GDP reaches some value, a larger GDP is associated with a larger support towards the education level.

- **Fertility Rate (FertRate):** The children per woman ratio influences family resource allocation for education. Higher fertility rate which implies a greater number of children per woman is associated with lower literacy levels as educating more children is more difficult than focusing on only one or two.

- **Share of Education in Government Expenditure (EducexpGDP):** This highlights the priority governments place on education, which is key to improving education levels.

- **Urbanization Rate (UrbanRatio):** Urbanization often correlates with better access to schools and resources. The larger is thelarger the urbanization of a population the higher is the chance that everyone gets access to education.
- **Democracy Index (Dem):** Democratic governance often correlates with stronger investment in public services, including education. It can be illustrated through the examples of corrupted countries with little or no democracy that have only a minor part of their population educated.
- **Average Years of Schooling (SchoolYears):** This variable measures the total time people spend in formal education and provides insights into the quality and reach of the education system. It is expected that the more grades on average students take in school, the higher the literacy rate will be.
- **Continent Dummies (Africa, Asia, Europe, South America, Oceania):** These account for regional differences in education systems, culture, and socio-economic conditions. North America is used as the reference group for comparison.

All these variables will describe factors that influence the Literacy Rate. Further, for regression using Stata, the dataset data for 110 countries were manually cleaned and compiled into an Excel spreadsheet.

3.1 Model

The full model consists of 16 explanatory variables:

$$\begin{aligned}\text{LitRate} = \beta_0 &+ \beta_{\text{GERsec}} \cdot \text{GERsec} + \beta_{\text{GERtert}} \cdot \text{GERtert} + \beta_{\text{PopDens}} \cdot \text{PopDens} \\ &+ \beta_{\text{Internet}} \cdot \text{Internet} + \beta_{\text{GDPCth}} \cdot \text{GDPCth} + \beta_{\text{GDPCthsq}} \cdot \text{GDPCthsq} \\ &+ \beta_{\text{FertRate}} \cdot \text{FertRate} + \beta_{\text{EducexpGDP}} \cdot \text{EducexpGDP} + \beta_{\text{UrbanRatio}} \cdot \text{UrbanRatio} \\ &+ \beta_{\text{Dem}} \cdot \text{Dem} + \beta_{\text{SchoolYears}} \cdot \text{SchoolYears} + \beta_{\text{Africa}} \cdot \text{Africa} + \beta_{\text{Europe}} \cdot \text{Europe} \\ &+ \beta_{\text{Asia}} \cdot \text{Asia} + \beta_{\text{SAmerica}} \cdot \text{SAmerica} + \beta_{\text{Oceania}} \cdot \text{Oceania}\end{aligned}$$

The following table describes and shows the main characteristics of each variable:

Variable	Variable Level	Obs	Mean	Std. dev.	Min	Max
LitRate	Literacy rate in %	177	87.64541	17.31866	27.28	100
GERsec	Gross enrollment ratio in secondary education	130	92.77686	23.4327	33.72491	144.8525
GERtert	Gross enrollment ratio in tertiary education	141	50.12097	29.99563	2.702	150.2018
PopDens	Population density per km^2	191	303.3768	1472.462	2.173997	18689.54

Internet	Percentage of people using Internet	191	67.10506	26.60163	2.004049	100
GDPCth	Gross Domestic Product per Capita in th. dol.	174	21.74438	22.53505	.3526	117.747
FertRate	Fertility rate : average number of children per woman	184	2.61038	1.175139	1.24	6.86
EducexpGDP	Education expense per GDP	162	4.428025	1.856813	1.2	15.43
UrbanRatio	Ratio between Urban and Total population in %	194	60.04397	22.98393	13.582	100
Dem	Democracy index	161	5.288634	2.380014	.32	9.81
SchoolYears	Average Years of Schooling	192	8.999439	3.222877	1.341352	14.25594
Africa	Africa	194	.2628866	.4413403	0	1
Europe	Europe	194	.2474227	.4326309	0	1
Asia	Asia	194	.2371134	.4264128	0	1
SAmerica	South America	194	.0618557	.2415165	0	1
Oceania	Australia and Oceania	194	.0721649	.2594303	0	1

Table 1: Description of the variables

Units of measurements:

In most countries, GDP per capita is measured in thousands of dollars which is the reason for the coefficients in the model becoming small and even smaller when they are in quadratic terms (a lot of decimal zeros). Thus, for the sake of a more accurate model, this variable was divided by 1000 and by $1000^2 = 1000000$ as a quadratic term. Regarding other variables, they were left unchanged. The Literacy Rate, Internet Access and Urbanization Ratio are all measured as a percentage.

Quadratic term and dummy variables:

This model includes the quadratic term GDP per capita squared since there may be a turnover point where $GDPCth$ will positively affect the literacy rate after reaching this.

Dummy variable in the model is Continents as every country belongs to a single continent and North America was chosen as the reference since the literacy rate of countries in this continent is the highest among all other countries. It is assumed that the coefficient of other countries will be negative.

3.2 Pretests

This model contains 16 variables along with quadratic terms and dummy variables. Assuming the fact that the variance is a constant term there, it can be seen that

the overall F-statistics of the model equating 23.19 is quite large and thus, it proves that the assumption regarding coefficients of all variables being equal to zero can be rejected. Additionally, it is observed that the R-squared equals 0.8 while the adjusted R-squared equals 0.765 which means that 80 percent of the variance in the dependent variable can be explained, indicating a near perfect fit. It is demonstrated that parameters such as Enrollment into Tertiary Education as well as the Population Density, Urban Ratio, and Democracy Rate have a low t-value. The testing of a hypothesis that their coefficients are jointly insignificant will follow in the Results section.

Number of obs	=	110
F(16, 93)	=	23.41
Prob > F	=	0.0000
R-squared	=	0.8011
Adj R-squared	=	0.7669
Root MSE	=	6.5678

Source	SS	df	MS
Model	16160.415	16	1010.02594
Residual	4011.70802	93	43.1366454
Total	20172.123	109	185.065349

LitRate	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
GERsec	.1727798	.0596506	2.90	0.005	.0543256	.291234
GERtert	-.0209845	.0430017	-0.49	0.627	-.1063774	.0644084
PopDens	.0005566	.0010386	0.54	0.593	-.0015058	.002619
Internet	.0859028	.0650371	1.32	0.190	-.0432481	.2150537
GDPCth	-.3577367	.1505151	-2.38	0.020	-.6566299	-.0588435
GDPCthsq	.0018814	.001194	1.58	0.119	-.0004897	.0042524
FertRate	-.6838509	1.64438	-0.42	0.678	-3.949264	2.581562
EducexpGDP	-.9170286	.4603904	-1.99	0.049	-1.831273	-.0027844
UrbanRatio	.0591051	.0615583	0.96	0.339	-.0631376	.1813477
Dem	-.2992012	.4096959	-0.73	0.467	-1.112776	.5143737
SchoolYears	2.994846	.4256837	7.04	0.000	2.149523	3.840169
Africa	-11.14505	3.375751	-3.30	0.001	-17.84862	-4.441477
Europe	-2.820236	2.595146	-1.09	0.280	-7.973682	2.333209
Asia	-5.046345	2.553421	-1.98	0.051	-10.11693	.0242436
SAmerica	-1.9494	3.166163	-0.62	0.540	-8.236771	4.337972
Oceania	-3.64466	4.52131	-0.81	0.422	-12.62309	5.333765
_cons	55.17567	8.099136	6.81	0.000	39.09239	71.25895

Table 2: The Regression Model

4 Results and Discussion

4.1 Heteroskedasticity

To check whether the variance is constant or not, the Breusch-Pagan Test was used. The found residuals were squared up and regressed with x variables. Regress the residuals on X_i:

$$\hat{u}_i^2 = \gamma_0 + \gamma_1 X_i$$

$$H_0 : \gamma_1 = 0, \text{ which means that variance is constant}$$
$$H_1 : \text{Variance is not constant}$$

The overall F-statistic found after regressing squares of residuals is 3.68 with $Prob > F = 0.0000$ which is why the homoskedasticity hypothesis is rejected.

Regressing model using Robust Standard Error:

Since the model with heteroskedasticity violates the F and t-tests it is impossible to use the standard error. After the Robust Standard Error was employed, the R-squared of the model has increased and now the F and t-tests can be conducted without issues.

Number of obs	=	110
F(16, 93)	=	14.94
Prob > F	=	0.0000
R-squared	=	0.8011
Root MSE	=	6.5678

LitRate	Coefficient	Robust SE	t	P>|t|	[95% conf.	interval]
GERsec	.1727798	.0562087	3.07	0.003	.0611605	.2843991
GERtert	-.0209845	.0314974	-0.67	0.507	-.0835321	.0415631
PopDens	.0005566	.0006896	0.81	0.422	-.0008128	.001926
Internet	.0859028	.0682829	1.26	0.212	-.0496936	.2214992
GDPCth	-.3577367	.1526937	-2.34	0.021	-.6609562	-.0545173
GDPCthsq	.0018814	.0009127	2.06	0.042	.0000688	.0036939
FertRate	-.6838509	2.136488	-0.32	0.750	-4.926493	3.558791
EducexpGDP	-.9170286	.4347701	-2.11	0.038	-1.780396	-.0536613
UrbanRatio	.0591051	.0606806	0.97	0.333	-.0613946	.1796048
Dem	-.2992012	.4501099	-0.66	0.508	-1.19303	.5946279
SchoolYears	2.994846	.5874958	5.10	0.000	1.828196	4.161496
Africa	-11.14505	5.59593	-1.99	0.049	-22.25746	-.0326436
Europe	-2.820236	3.653439	-0.77	0.442	-10.07524	4.434769
Asia	-5.046345	3.180984	-1.59	0.116	-11.36315	1.270459
SAmerica	-1.9494	2.686417	-0.73	0.470	-7.284091	3.385292
Oceania	-3.64466	3.957205	-0.92	0.359	-11.50288	4.213564
_cons	55.17567	9.79906	5.63	0.000	35.71668	74.63467

Table 3: The Regression Model using Robust Standard Error

The results of the regression were quite expected, except for some parameters. For example, the coefficients for *GERtert*, and *Dem* turned out to be negative, while the partial effect of *PopDens* was positive. This may be related to the fact that, in the past, low levels of democracy, weak education systems—especially higher education—and high population density hindered countries (often developing ones) from implementing effective policies to improve literacy. However, now these factors may either be insignificant or are already having the opposite effect. All of these factors have a higher t-statistic and will be analyzed jointly with 2 other variables in the next section.

4.2 Inferences

The observation is as follows: variables such as Gross Enrollment Ratio in Tertiary Education, Population Density, Fertility Rate, Democracy Index, and Urbanization Rate. They have low t-statistics and we hypothesize that all of the 5 coefficients are equal to zero.

$$H_0 : \beta_{\text{GERtert}} = 0,\ \beta_{\text{PopDens}} = 0,\ \beta_{\text{FertRate}} = 0,\ \beta_{\text{Dem}} = 0,\ \beta_{\text{UrbanRatio}} = 0$$
$$H_1 : \text{At least one of the coefficients is not zero}$$

As a result, F-statistics turned out to be 0.94 with $Prob > F = 0.4599$ which is why the hypothesis about all of the 5 coefficients being equal to zero is rejected.
Therefore, the reduced model is:

$$\begin{aligned}\text{LitRate} = \beta_0 &+ \beta_{\text{GERsec}} \cdot \text{GERsec} + \beta_{\text{Internet}} \cdot \text{Internet} + \beta_{\text{GDPCth}} \cdot \text{GDPCth} \\ &+ \beta_{\text{GDPCthsq}} \cdot \text{GDPCthsq} + \beta_{\text{EducexpGDP}} \cdot \text{EducexpGDP} \\ &+ \beta_{\text{SchoolYears}} \cdot \text{SchoolYears} + \beta_{\text{Africa}} \cdot \text{Africa} + \beta_{\text{Europe}} \cdot \text{Europe} \\ &+ \beta_{\text{Asia}} \cdot \text{Asia} + \beta_{\text{SAmerica}} \cdot \text{SAmerica} + \beta_{\text{Oceania}} \cdot \text{Oceania}\end{aligned}$$

Regressing the reduced regression model gives the results below:

Number of obs	=	116
F(11, 104)	=	16.33
Prob > F	=	0.0000
R-squared	=	0.7796
Root MSE	=	6.5678

LitRate	Coefficient	Robust SE	t	P>|t|	[95% conf.	interval]
GERsec	.1343853	.0431181	3.12	0.002	.0488805	.2198901
Internet	.1371818	.0672065	2.04	0.044	.0039088	.2704548
GDPCth	-.3264441	.117418	-2.78	0.006	-.5592883	-.0935998
GDPCthsq	.0017557	.0007538	2.33	0.022	.0002609	.0032505
EducexpGDP	-.8516507	.4265849	-2.00	0.048	-1.697585	-.0057168
SchoolYears	2.989655	.5646307	5.29	0.000	1.869971	4.109339
Africa	-11.32437	3.572174	-3.17	0.002	-18.40812	-4.240613
Europe	-4.065383	3.161522	-1.29	0.201	-10.3348	2.204033
Asia	-4.611158	2.814124	-1.64	0.104	-10.19167	.9693553

SAmerica	-1.929749	2.248872	-0.86	0.393	-6.389346	2.529848
Oceania	-4.889146	2.70291	-1.81	0.073	-10.24912	.4708269
_cons	54.19918	6.966633	7.78	0.000	40.38409	68.01428

Table 4: The Reduced Regression Model

The possible reasoning for the insignificance of those factors is that they do not affect the literacy rate directly. Tertiary education is insignificant since its existence already implies that people are educated. Population density may have little impact on literacy rates if other urban or rural factors, such as access to education or infrastructure, play a more dominant role. The Democracy Index might be insignificant because factors such as political stability or governance quality may matter more than the political system in determining literacy outcomes. An example of that is a high literacy rate observed during the existence of the Soviet Union - a country with an authoritarian regime but educated population. Population density may have little impact on literacy rates if other urban or rural factors, such as access to education or infrastructure, play a more dominant role. For instance, China or India are the countries with the most population and high population density but high literacy rates as well. Fertility rate might be insignificant in explaining literacy as it doesn't necessarily capture the educational opportunities or quality that directly affect literacy levels. Urbanization rate may be insignificant if urban-rural disparities in educational access are not strong enough to influence national literacy rates.

4.3 Analysis of quadratic terms

The hypothesis that $GDPCth$ and $GDPCthsq$ do not affect the literacy rate of the population was tested:

$$H_0 : \beta_{\text{GDPCth}} = 0,\ \beta_{\text{GDPCthsq}} = 0$$
$$H_1 : \text{At least one of the coefficients is not zero}$$

As a result, the F-statistic is equal 3.14 with $Prob > F = 0.0480$, which means that the null hypothesis is rejected at a 95% confidence level.
Also, the hypothesis that $GDPCthsq$ is a linear, not the quadratic function is tested:

$$H_0 : \beta_{\text{GDPCthsq}} = 0$$
$$H_1 : \beta_{\text{GDPCthsq}} \neq 0$$

As a result, the F-statistic is equal 4.25 with $Prob > F = 0.0421$, which means that the null hypothesis is rejected at a 95% confidence level.

Moreover, the turnover point was found by the formula:

$$Turn.p. = \left| \frac{\beta_{\text{GDPCth}}}{2 \cdot \beta_{\text{GDPCthsq}}} \right|$$

In the unrestricted and restricted regression line, the turnover point of GDP per capita is estimated to be 93 000 dollars. Since in both model, the coefficient in $GDPCth$ is negative and in $GDPCthsq$ is positive, this means that before reaching

the turnover value, GDPC negatively affects the literacy rate, and after reaching this, the impact changes into a positive one.

4.4 Analysis of dummy variables

Analyzing the partial effects of each continent, we can see that in North America there are countries with more literacy levels since all of the coefficients of other continents are negative. For example in the unrestricted model, compared with Europe, the average literacy rate of North American Countries is 2.8% larger. In both restricted and unrestricted models it can be seen that the coefficients are negative which implies that, compared to North American countries, all other countries in the rest of the world have lower literacy rates.

5 Conclusion

The main objective of this paper was to examine the factors affecting literacy rates across countries using data from 110 countries and an econometric model. Literacy is critical for economic growth, reducing inequality, and improving living conditions, so understanding what drives it is important for designing effective policies. This study focused on identifying the key variables affecting literacy and analyzing their significance. The results showed that factors such as gross secondary school enrollment, access to the internet, GDP per capita public expenditure on education, and school years play a significant role in improving the literacy rates of a population. These results highlight how access to education, the use of technology, and investment in education systems can positively influence literacy outcomes. The analysis also showed that GDP per capita has a non-linear relationship with literacy, where its positive effects become visible only after reaching a certain level. In addition, the study found regional differences. Countries in North America show higher average literacy rates compared to other continents such as Africa, Europe, Australia and Oceania, South America, and Asia. In this analysis, it was tested that 5 variables such as Gross Enrollment Ratios in Tertiary Education, Population Density, Fertility Rate, Democracy Index, and Urbanization Rate are jointly insignificant. Therefore, these variables can be removed from the model, leaving only those variables that are really important for the result. However, this paper has some limitations. One of the main problems was the exclusion of some variables such as pupil/teacher ratio. These variables were removed because not enough data were found. Including these missing variables in our model could provide a deeper understanding of what influences the literacy of the population. Another limitation is that the study used cross-sectional data, which provides an estimate of one specific period of time and does not show how literacy changes over time or as a result of the implementation of any policy in the country. To improve future studies, it would be useful to include more complete data sets with missing variables. Moreover, it is important to compare different time periods to get a full picture of the situation. Future research could also focus on new issues, for example, how digital education or global issues such as migration and climate change affect literacy rates, which shows great potential for further study.

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